

Answers: Logic Drill #5

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Section A

1. In the proof step “If $a = b$, then $a(a - b) = (a - b)^2 \implies a = a - b$ ”, what operation was invalidly performed?

Answer: Division by zero

Method: Since $a = b$, we have $a - b = 0$. Dividing both sides by $(a - b)$ is division by zero, which is algebraically undefined.

Investigate further: Why is division by zero forbidden in field axioms?

2. True or false: “ $\sqrt{x^2} = x$ for all real numbers x .”

Answer: False

Method: By definition, $\sqrt{x^2} = |x|$ for all real x . For negative x (e.g. $x = -3$), $\sqrt{(-3)^2} = \sqrt{9} = 3 \neq -3$.

Investigate further: For which real numbers x is $\sqrt{x^2} = x$ true?

3. What fallacy is committed when one assumes the conclusion P in order to prove P ?

Answer: Begging the question (*petitio principii* / circular reasoning)

Method: Assuming the very claim P that needs to be established creates a circular argument where the truth of P relies on P itself.

Investigate further: How does circular reasoning differ from proof by contradiction?

4. A student writes: “ $x^2 > 9 \implies x > 3$.” What case did the student miss?

Answer: The case $x < -3$

Method: $x^2 > 9 \iff |x| > 3 \iff x > 3 \text{ or } x < -3$. The student omitted the negative interval.

Investigate further: Solve $x^2 < 9$ over the real numbers.

5. True or false: “If $ac \equiv bc \pmod{m}$, then $a \equiv b \pmod{m}$ for all non-zero integers c, m .”

Answer: False

Method: Cancellation in modular arithmetic requires $\gcd(c, m) = 1$. For example, $2 \times 3 \equiv 4 \times 3 \pmod{6}$ ($6 \equiv 12 \equiv 0 \pmod{6}$), but $2 \not\equiv 4 \pmod{6}$.

Investigate further: What is the general cancellation rule for $ac \equiv bc \pmod{m}$?

6. In the deduction “If $x^2 = 4$, then $x = 2$ ”, which direction of logical implication was omitted?

Answer: The possibility $x = -2$

Method: $x^2 = 4 \implies x = 2$ or $x = -2$. Omitting $x = -2$ assumes a one-sided implication that ignores the negative root.

Investigate further: Is $x = 2 \implies x^2 = 4$ a true implication?

7. Identify the error in: “Since $-1 = (-1)^1 = (-1)^{2/2} = ((-1)^2)^{1/2} = 1^{1/2} = 1$.”

Answer: The exponent rule $(x^a)^b = x^{ab}$ fails when the base is negative and exponents involve fractions.

Method: $((-1)^2)^{1/2} = 1^{1/2} = 1$, but $(-1)^{2/2} = (-1)^1 = -1$. The law $(x^m)^n = x^{mn}$ requires a positive base when non-integer powers are used.

Investigate further: Why must non-integer exponents be defined carefully over negative real bases?

8. True or false: “Proving that the converse of a statement is true proves the original statement.”

Answer: False

Method: A statement $P \implies Q$ and its converse $Q \implies P$ are logically independent. The truth of $Q \implies P$ implies nothing about the truth of $P \implies Q$.

Investigate further: Which related statement always shares the exact same truth value as $P \implies Q$?

9. What is invalid about dividing both sides of $x^2 - x = 0$ by x to conclude $x = 1$?

Answer: It loses the root $x = 0$ (and divides by zero if $x = 0$).

Method: Dividing by x implicitly assumes $x \neq 0$, thereby discarding the valid solution $x = 0$. The correct approach is factorising $x(x - 1) = 0 \implies x = 0$ or $x = 1$.

Investigate further: When is dividing an equation by an algebraic expression valid?

10. A student claims: “ $0 \times 1 = 0 \times 2$, so cancelling 0 gives $1 = 2$.” Which field axiom is violated?

Answer: Existence of multiplicative inverse (only non-zero elements have multiplicative inverses).

Method: In a field, every element $a \neq 0$ has a multiplicative inverse a^{-1} . Zero has no multiplicative inverse, so cancellation by 0 is undefined.

Investigate further: Prove from field axioms that $0 \times x = 0$ for all x .

Section B

1. Consider the attempt to prove $2 = 1$:

Line 1: Let $a = b$.

Line 2: Then $a^2 = ab$.

Line 3: Subtract b^2 : $a^2 - b^2 = ab - b^2$.

Line 4: Factorise: $(a - b)(a + b) = b(a - b)$.

Line 5: Divide by $(a - b)$: $a + b = b$.

Line 6: Substitute $a = b$: $b + b = b \implies 2b = b$.

Line 7: Divide by b : $2 = 1$.

Which line contains the first invalid step?

Answer: Line 5

Method: Since $a = b$ (Line 1), $a - b = 0$. In Line 5, dividing by $(a - b)$ is division by zero, which is invalid.

Investigate further: Why does division by zero allow proving any false statement?

2. Consider the attempt to prove $1 = -1$:

Line 1: $-1 = \frac{-1}{1}$.

Line 2: Taking square roots: $\sqrt{-1} = \sqrt{\frac{-1}{1}}$.

Line 3: $\sqrt{-1} = \frac{\sqrt{-1}}{\sqrt{1}}$.

Line 4: Also $\sqrt{-1} = \sqrt{\frac{1}{-1}} = \frac{\sqrt{1}}{\sqrt{-1}}$.

Line 5: Equating gives $\frac{\sqrt{-1}}{1} = \frac{1}{\sqrt{-1}}$.

Line 6: Cross-multiplying gives $(\sqrt{-1})^2 = 1 \times 1 \implies -1 = 1$.

Which line contains the first invalid manipulation of square roots?

Answer: Line 3 (or Line 4)

Method: The radical property $\sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}}$ holds only when $x \geq 0$ and $y > 0$. Applying it to negative numbers introduces multivalued complex branch conflicts.

Investigate further: What is the principal value of $\sqrt{-1}$ in complex numbers?

3. Consider the argument:

Line 1: We want to show $x^2 + 1 > 0$ for all real x .

Line 2: Assume $x^2 + 1 > 0$.

Line 3: Subtract 1: $x^2 > -1$.

Line 4: Since $x^2 \geq 0$ for all real x , $x^2 > -1$ is true.

Line 5: Therefore $x^2 + 1 > 0$ is proved.

What is the logical defect in this structure?

Answer: Begging the question / working backwards without bi-implications

Method: Line 2 assumes the conclusion $x^2 + 1 > 0$ to derive a true statement $x^2 > -1$. Unless every step is explicitly shown to be reversible ($\iff$), starting from the desired conclusion does not prove it.

Investigate further: How should this proof be written correctly from first principles?

4. Consider the proof attempt for: “For all real x , if $x^2 > 4$, then $x > 2$.”

Line 1: Suppose $x^2 > 4$.

Line 2: Take square roots: $\sqrt{x^2} > \sqrt{4}$.

Line 3: Simplify: $x > 2$.

Line 4: Hence the claim is proved.

Which line contains the invalid simplification, and what should it be?

Answer: Line 3; it should be $|x| > 2$ (which means $x > 2$ or $x < -2$).

Method: $\sqrt{x^2} = |x|$, so Line 2 gives $|x| > 2$. Simplifying $|x| > 2$ yields $x > 2$ or $x < -2$. Line 3 incorrectly drops the $x < -2$ branch.

Investigate further: Provide a counterexample to the claim “if $x^2 > 4$ then $x > 2$ ”.

5. Consider the argument:

Line 1: Let $S = 1 + 2 + 4 + 8 + 16 + \dots$

Line 2: Then $2S = 2 + 4 + 8 + 16 + \dots$

Line 3: Subtracting gives $S - 2S = 1 \implies -S = 1 \implies S = -1$.

Line 4: Therefore $1 + 2 + 4 + 8 + \dots = -1$.

Which line relies on an invalid assumption about infinite series?

Answer: Line 1 (and Line 3)

Method: The geometric series $1 + 2 + 4 + 8 + \dots$ has ratio $r = 2 > 1$, so it diverges to $+\infty$. Assigning a real variable S to a divergent series and performing linear algebra on it is invalid.

Investigate further: In p -adic analysis or analytic continuation, does $1 + 2 + 4 + 8 + \dots = -1$ have a valid formal meaning?

6. Consider the attempt to prove every integer is even:

Line 1: Let $P(n)$ be the statement “ n is even”.

Line 2: Assume $P(k)$ is true, so k is even, i.e., $k = 2m$.

Line 3: Then $k + 2 = 2m + 2 = 2(m + 1)$, which is even, so $P(k + 2)$ is true.

Line 4: Since $P(0)$ is true ($0 = 2 \times 0$), $P(n)$ holds for all even n .

Line 5: Therefore all integers are even.

Which line makes an unjustified logical leap?

Answer: Line 5

Method: Lines 1–4 correctly prove by induction that $P(n)$ holds for all *even* non-negative integers. Line 5 falsely leaps to concluding that *all* integers (odd and negative) are even.

Investigate further: Why did the step $k \rightarrow k + 2$ skip odd numbers?

7. Consider:

Line 1: $a < b$.

Line 2: Multiply both sides by c : $ac < bc$.

Line 3: Subtract ac : $0 < bc - ac = c(b - a)$.

Line 4: Divide by $b - a$ (which is > 0): $0 < c \implies c > 0$.

What condition on c was implicitly assumed in Line 2?

Answer: That $c > 0$

Method: Multiplying an inequality by c preserves the inequality sign ($ac < bc$) if and only if $c > 0$. If $c < 0$, the inequality sign flips ($ac > bc$).

Investigate further: What happens when $c = 0$ in Line 2?

8. Consider the argument:

Line 1: To prove $a = b$, it suffices to show $a^2 = b^2$.

Line 2: Take $a = 3$ and $b = -3$.

Line 3: $3^2 = (-3)^2 = 9$.

Line 4: Therefore $3 = -3$.

Which line contains a false logical equivalence?

Answer: Line 1

Method: $a^2 = b^2 \iff a = b \text{ or } a = -b$. $a^2 = b^2$ is necessary for $a = b$, but not sufficient because a could equal $-b$.

Investigate further: Is $a = b \implies a^2 = b^2$ a valid implication?

9. Consider the proof attempt for $\sqrt{2} + \sqrt{3} < \sqrt{10}$:

Line 1: Square both sides: $(\sqrt{2} + \sqrt{3})^2 < 10$.

Line 2: Expand: $2 + 2\sqrt{6} + 3 < 10 \implies 5 + 2\sqrt{6} < 10$.

Line 3: Subtract 5: $2\sqrt{6} < 5$.

Line 4: Square both sides: $4 \times 6 < 25 \implies 24 < 25$.

Line 5: Since $24 < 25$ is true, the original inequality holds.

Why is this argument incomplete as written?

Answer: It works backwards without explicitly stating that every squaring step is reversible for positive quantities.

Method: Squaring $A < B \implies A^2 < B^2$ is reversible ($\iff$) ONLY when both $A > 0$ and $B > 0$. Since all expressions are positive, the steps are reversible, but the proof must state this reversibility explicitly to be complete.

Investigate further: Re-write this proof starting from $24 < 25$ and working forward to $\sqrt{2} + \sqrt{3} < \sqrt{10}$.

10. Consider the argument:

Line 1: $1^2 = 1$.

Line 2: $2^2 = 1 + 3$.

Line 3: $3^2 = 1 + 3 + 5$.

Line 4: By pattern recognition, n^2 is the sum of the first n odd positive integers.

Line 5: Therefore the theorem is proved for all $n \in \mathbb{Z}^+$.

Why is pattern recognition up to $n = 3$ not a rigorous proof?

Answer: Checking a finite number of examples does not prove a universal statement for all $n \in \mathbb{Z}^+$.

Method: Empirical verification for $n = 1, 2, 3$ demonstrates a pattern but does not constitute a proof for infinitely many integers. A complete proof requires mathematical induction or algebraic sum formula $\sum_{k=1}^n (2k-1) = n^2$.

Investigate further: Prove $\sum_{k=1}^n (2k-1) = n^2$ using mathematical induction.

Section C

1. A student attempts to prove: “For all real x , if $x^3 > x$, then $x > 1$.” (after TMUA 2021 Paper 2 Q5)

Answer: A) Line 3 is invalid when $x < 0$, and Line 4 misses the possibility $x < -1$.

Method: In Line 3, dividing $x(x^2 - 1) > 0$ by x requires knowing the sign of x . If $x < 0$, dividing by x flips the inequality to $x^2 - 1 < 0 \implies -1 < x < 0$. If $x > 0$, dividing by x gives $x^2 - 1 > 0 \implies x > 1$. Furthermore, $x^2 > 1$ gives $x > 1$ or $x < -1$. Thus $x^3 > x$ is satisfied for $x \in (-1, 0) \cup (1, \infty)$.

Investigate further: What is the correct solution set to $x^3 > x$?

2. Consider the statement: “If n is an integer such that n^2 is divisible by 6, then n is divisible by

6.” (after TMUA 2018 Paper 2 Q6)

Answer: B) No, Line 3 assumes what it tries to prove by asserting $k = 6m^2$.

Method: Line 3 asserts $k = 6m^2$ without proof. Knowing $\sqrt{6k} \in \mathbb{Z}$ means $6k$ is a square, so $6k = 36m^2 \implies k = 6m^2$. But assuming $k = 6m^2$ directly assumes that $6 \mid k$, which assumes that $36 \mid n^2 \implies 6 \mid n$, circular reasoning!

Investigate further: How do you correctly prove $6 \mid n^2 \implies 6 \mid n$ using prime factorisation?

3. A student writes a proof of the claim: “For all positive integers n , $n^2 + 5n + 6$ is even.”

Answer: A) The proof is completely valid and correct.

Method: Line 1 factorises correctly to $(n+2)(n+3)$. Line 2 correctly notes that $n+2$ and $n+3$ are consecutive integers, so one is even and one is odd. Line 3 correctly uses the fact that even $\times$ odd = even. The proof is entirely sound.

Investigate further: Can you also prove $n^2 + 5n + 6$ is even by testing n even and n odd separately?

4. A student attempts to prove: “For all real x , $x^2 + 4x + 5 > 0$.”

Answer: A) The proof is completely valid and correct.

Method: Line 1 completes the square: $x^2 + 4x + 5 = (x+2)^2 + 1$. Line 2 correctly uses $(x+2)^2 \geq 0 \implies (x+2)^2 + 1 \geq 1 > 0$. Line 3 correctly concludes $x^2 + 4x + 5 > 0$. Every step is rigorous and valid.

Investigate further: What is the minimum value of $x^2 + 4x + 5$, and for which x does it occur?

5. A student attempts to prove: “If $a > b > 0$, then $a^2 > b^2$.”

Answer: A) The proof is completely valid and correct.

Method: Since $a > 0$ and $b > 0$, multiplying $a > b$ by $a > 0$ preserves inequality: $a^2 > ab$. Multiplying $a > b$ by $b > 0$ preserves inequality: $ab > b^2$. Transitivity gives $a^2 > ab > b^2 \implies a^2 > b^2$. All steps are valid.

Investigate further: Why does this proof fail if $b < 0$?

6. Consider the attempt to prove: “If x^2 is irrational, then x is irrational.”

Answer: A) The proof is completely valid and correct.

Method: The contrapositive of $P \implies Q$ ($x^2 \notin \mathbb{Q} \implies x \notin \mathbb{Q}$) is $\neg Q \implies \neg P$ ($x \in \mathbb{Q} \implies x^2 \in \mathbb{Q}$). If $x = p/q \in \mathbb{Q}$, then $x^2 = p^2/q^2 \in \mathbb{Q}$. Since the contrapositive is true, the original statement is true.

Investigate further: Is the converse “if x is irrational then x^2 is irrational” true?

7. A student attempts to prove: “For all real $x \neq 1$, $\frac{x^2-1}{x-1} = x+1$.”

Answer: A) The proof is completely valid and correct.

Method: For $x \neq 1$, $x-1 \neq 0$, so dividing numerator and denominator by $(x-1)$ is division by a non-zero number, which is valid. $\frac{(x-1)(x+1)}{x-1} = x+1$.

Investigate further: What happens to the expression at $x = 1$?

8. Consider the claim: “For all integers n , $n^3 - n$ is divisible by 3.”

Answer: A) The proof is completely valid and correct.

Method: $n^3 - n = (n - 1)n(n + 1)$ is the product of three consecutive integers. In any three consecutive integers, exactly one is a multiple of 3, so their product is divisible by 3. The argument holds for all $n \in \mathbb{Z}$.

Investigate further: Is $n^3 - n$ also always divisible by 6?

Section D

1. Consider the following attempt to prove: “For all real numbers x , if $x^2 - 5x + 6 = 0$, then $x = 2$.” (after TMUA 2016 Paper 2 Q11)

Answer: C) Line 4 does not follow because $x = 3$ is also a solution, so $x^2 - 5x + 6 = 0$ does not imply $x = 2$ exclusively.

Method: Line 1 factorises correctly to $(x - 2)(x - 3) = 0$. Line 2 correctly states $x = 2$ or $x = 3$. Line 3 correctly verifies that $x = 2$ is a solution. However, Line 4 falsely concludes $x = 2$ exclusively, ignoring $x = 3$. Since $x = 3$ is also a valid root, $x^2 - 5x + 6 = 0$ does not imply $x = 2$ (the statement is false for $x = 3$).

Investigate further: What is the correct logical relationship between $x^2 - 5x + 6 = 0$ and $x \in \{2, 3\}$?

2. A student attempts to prove: “All horses are the same color” using induction.

Answer: B) The transition from $k = 1$ to $k = 2$ fails because the two subsets $\{h_1\}$ and $\{h_2\}$ do not overlap.

Method: For $k = 1$, the two subsets are $\{h_1\}$ and $\{h_2\}$. These two 1-element subsets share no common element (h_2 is not in $\{h_1\}$). Thus, the overlap argument that forces h_1 and h_2 to share the color of the overlapping element fails at $k = 1 \rightarrow k = 2$.

Investigate further: Why does the overlap argument work for $k \geq 2 \rightarrow k + 3$ but fail at $k = 1$?

3. Consider the attempt to prove: “If n^2 is divisible by 3, then n is divisible by 3.” (after TMUA 2017 Paper 2 Q11)

Answer: A) The proof is completely valid and rigorous.

Method: Line 1 correctly states the contrapositive: $3 \nmid n \implies 3 \nmid n^2$. Line 2 covers all cases for $3 \nmid n$: $n \equiv 1 \pmod{3}$ or $n \equiv 2 \pmod{3}$. Line 3 shows $(3k + 1)^2 = 3(3k^2 + 2k) + 1 \not\equiv 0 \pmod{3}$. Line 4 shows $(3k + 2)^2 = 3(3k^2 + 4k + 1) + 1 \not\equiv 0 \pmod{3}$. Line 5 correctly concludes the contrapositive holds, proving the theorem.

Investigate further: Why is proving by contrapositive easier than direct proof here?

4. A student writes the following proof that $1 = 0$:

Answer: C) Line 5: division by $x - 1$ is division by zero since $x = 1$.

Method: In Line 1, $x = 1$, so $x - 1 = 0$. In Line 5, dividing both sides of $(x - 1)(x + 1) = x - 1$ by $x - 1$ is dividing by zero, which is algebraically invalid and produces the false identity $2 = 1$.

┆ **Investigate further:** At which line does the proof first depart from valid field operations?

5. Consider the statement: “For all real numbers x, y , if $x + y > 0$, then $x > 0$ or $y > 0$.” (after TMUA 2020 Paper 2 Q4)

Answer: A) The proof is completely valid and correct.

Method: The contrapositive of $P \implies (Q \vee R)$ is $\neg(Q \vee R) \implies \neg P$, which by De Morgan’s law is $(\neg Q \wedge \neg R) \implies \neg P$. Here $\neg Q \wedge \neg R$ is $x \leq 0 \wedge y \leq 0$, and $\neg P$ is $x + y \leq 0$. Adding $x \leq 0$ and $y \leq 0$ gives $x + y \leq 0$. Since the contrapositive is true, the original statement is true.

┆ **Investigate further:** Is the converse “if $x > 0$ or $y > 0$, then $x + y > 0$ ” true?